

DS	Session	Linked List	Question Information	Level 1	Challenge 31
Question description Professor Shiva decided to conduct an industrial visit for final year students, but he set a condition that if students received a passing grade in the surprise test, they would be eligible to go on the industrial visit. He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test. Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked. For example if the given Linked List is 5->10->15->10->25 and delete after 10 then the Linked List becomes 5->15->25. Constraints 1 < N < 100 1 < D < 1000 Input Format First line contains the number of datas- N. Second line contains N integers(the given linked list). Next line indicates the node data D that has to be deleted. Output Format Single line represents the linked list after required elements deleted.					
Logical Test Cases Test Case 1 INPUT (STDIN) 6 7 9 8 8 4 2 8 EXPECTED OUTPUT Linked List:->7->9->4->2 Test Case 2 INPUT (STDIN) 10 5 3 7 2 7 9 8 8 4 2 2 EXPECTED OUTPUT Linked List:->5->3->7->7->9->8->8->4					

▼ Mandatory Test Cases

Test Case 1

KEYWORD

`struct node`

Test Case 2

KEYWORD

`node *next;`

Test Case 3

KEYWORD

`void create()`

Test Case 4

KEYWORD

`p2=p2->next;`

Test Case 5

KEYWORD

`void del()`

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

280

Test Case 3

NLOC

65

LinkedList1.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *start = NULL;
10
11 void create(int val) {
12     struct node *temp = (struct node*)malloc(sizeof(struct node));
13     temp->data = val;
14     temp->next = NULL;
15     if (start == NULL) {
16         start = temp;
17     } else {
18         struct node *p = start;
19         while (p->next != NULL) {
20             p = p->next;
21         }
22         p->next = temp;
23     }
24 }
25
```

```
26 void del(int key) {
27     while (start != NULL && start->data == key) {
28         struct node *temp = start;
29         start = start->next;
30         free(temp);
31     }
32     struct node *p2 = start;
33     while (p2 != NULL && p2->next != NULL) {
34         if (p2->next->data == key) {
```

main | CH.SC.U4CSE25225 | LinkedList1.c

Ln 68, Col 1 Spaces: 4 UTF-8 CRLF C

```
36         p2->next = temp->next;
37         free(temp);
38     } else {
39         p2 = p2->next;
40     }
```

```
41     }
42 }
43
44 void display() {
45     printf("Linked List:");
46     struct node *p = start;
47     while (p != NULL) {
48         printf("->%d", p->data);
49         p = p->next;
50     }
51     printf("\n");
52 }
53
54 int main() {
```

```
55  int n;
56  if (scanf("%d", &n) != 1) return 0;
57  for (int i = 0; i < n; i++) {
58      int val;
59      scanf("%d", &val);
60      create(val);
61  }
62  int key;
63  scanf("%d", &key);
64  del(key);
65  display();
66  return 0;
67}
68
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
6
```

```
7 9 8 8 4 2
```

```
8
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList1.exe
```

```
Linked List:->7->9->4->2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
10
```

```
5 3 7 2 7 9 8 8 4 2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList1.exe
```

```
Linked List:->5->3->7->7->9->8->8->4
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```